1 Derivatives

The derivative of a function f at x_0 is defined as the limit

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h}.$$

It measures the instantaneous rate of change. Differentiability implies continuity, but not the converse.

1.1 Chain rule

For composite functions, $(f \circ g)'(x) = f'(g(x)) \cdot g'(x)$.

2 Integration

The fundamental theorem of calculus links differentiation and integration:

$$\int_a^b f'(x)\,dx = f(b) - f(a).$$

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